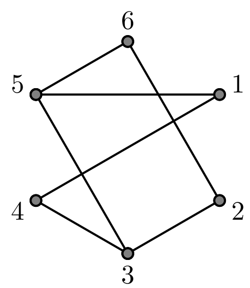
Math 252 C
Spring 2019

Name ______________________ Section: A B

Final Exam

1 Answer the following questions about the graph below.



a) What is an Euler circuit?

b) Does the graph have an Euler circuit? How do you know?

c) Does the graph have an Euler path that is not a circuit? How do you know?

d) Is the graph planar? How do you know?

e) Is the graph bipartite? How do you know?

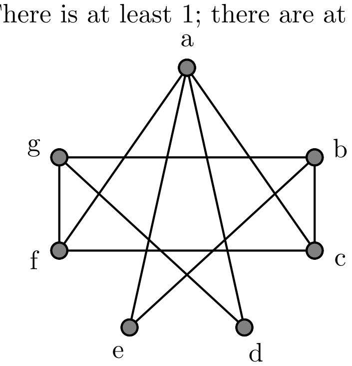
2 Draw a simple graph with 6 vertices, each of which has degree 4.

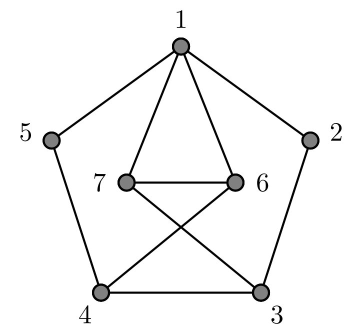
3 Explain why it is impossible to draw a simple graph with 7 vertices, each of degree 3.

4 Give an example of a graph that is **not a planar graph** and explain how you know it is not planar. **Do not use Kuratowski's Theorem.**

5 How many isomorphisms are there between the following two graphs? **Explain.**

(If you do not know exactly how many there are, tell me as much as you know about how many there are. Examples: There is at least 1; there are at most 3, etc.)





6 Show how to compute $C(13, 3) = \binom{12}{4}$ without using a calculator.

7 $\mathrm{P}(A) = 0.3$, $\mathrm{P}(B) = 0.5$, $\mathrm{P}(A \cap B) = 0.2$. Use this to answer the following questions. Be sure to show your work.

a) What is $\mathrm{P}(A \cup B)$?

b) What is $\mathrm{P}(A \mid B)$?

c) Are A and B mutually exclusive? (Explain your answer.)

d) Are A and B independent? (Explain your answer.)

8 COINS AND CARDS.

a) If I flip a coin 6 times, what is the probability of getting the same result each of the six times?

b) If I deal six cards from a standard 52 card deck, what is the probability that they will all be the same color? (The deck has 26 red cards and 26 black cards.)

9 4's on Four 4-Sided Dice. Let X be the number of 4's when rolling four 4-sided dice.

a) What is $\mathrm{P}(X = 0)$?

b) What is $\mathrm{P}(X = 2)$?

c) What is the expected value of X? (This is denoted $\mathrm{E}(X)$.)

10 Sam collects commemorative stamps. Among his favorites are Maria Goeppert Mayer stamps  and the new cowaphant stamps. 

On Sam's desk are four envelopes with stamps in them. The table below shows how many of each kind of stamp are in each envelope.

envelope	 Maria Goeppert Mayer	 cowaphant
A	1	4
B	3	7
C	10	10
D	18	2

Sam randomly selects an envelope (each envelope equally likely) and then randomly selects a stamp from within that envelope (each stamp in the envelope equally likely).

a) What is the probability that Sam chooses a cowaphant stamp?

b) If Sam chooses a cowaphant stamp, what is the probability that the envelope was envelope A?

11 Dice.

a) If you roll three dice (each with six sides numbered 1, 2, 3, 4, 5, and 6), what is the probability that the three numbers rolled will all be the same?

b) If you roll three dice (each with six sides numbered 1, 2, 3, 4, 5, and 6), what is the probability that the three numbers rolled will be distinct? (no repetitions)

12 What is the output of the following python code?

```
import re as re
re.findall(r'b+a*[cd]*', 'abcdabcddcbaabbccddccbbaa')
```

13 Let L be the langauge consisting of all binary strings that end with at least two 0's followed by at least one 1. Examples: $10011 \in L$, $0010 \notin L$, $01011 \notin L$.

a) Write a **regular expression** for L.

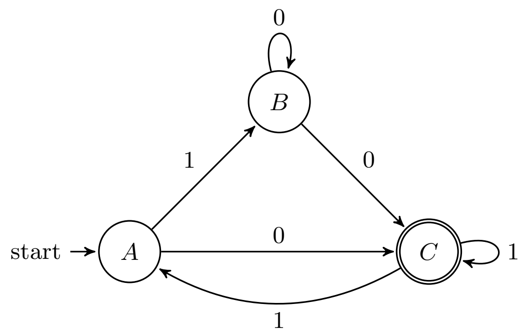
b) Construct a **DFA** that recognizes L.

14 Answer **ONE** of the following (your choice):

a) Give an example of a language that is **not regular**, and explain how you know the language is not regular.

b) Give an example of a language that is **not Turing computable** and explain how you know the language is not Turing computable.

15 Below is the state diagram for an **NFA** N.



a) Which of the following strings are accepted by N? (Simply write "accept" or "reject" after each.)

- 101
- 0100
- 0101
- 0011
- 1001
- 01100

b) Design a **DFA** that accepts the same language as N using a general method that works for all NFAs.

c) Construct a **regular grammar** for this language using a **general method** that works for all NFAs.

16 Below is the instruction set for a Turing machine M with start state `s0` and final state (i.e., accepting state) `s4`. In each row, the columns are in the order state, symbol, symbol, direction, state.

```
s0  0  X  R  s2
s0  1  X  R  s1
s1  0  Y  R  s2
s1  1  Y  R  s3
s2  0  Z  L  s4
s2  1  Z  R  s3
s3  0  0  R  s3
s3  1  1  L  s4
```

Fill out the table below:

input string	Does M accept? (Y or N)	tape contents upon halting (or DNH for does not halt)	last state (or DNH for does not halt)
0011			
1101			
0100			

This space available for extra work, cowaphant art, etc.

10

```
sum(.25 * c(4/5, 7/10, 1/2, 1/10))

## [1] 0.525

sum(.25 * c(4/5, 7/10, 1/2, 1/10)) %>% fractions()

## [1] 21/40

.25 * c(4/5, 7/10, 1/2, 1/10) / sum(.25 * c(4/5, 7/10, 1/2, 1/10))

## [1] 0.38095238 0.33333333 0.23809524 0.04761905

.25 * c(4/5, 7/10, 1/2, 1/10) / sum(.25 * c(4/5, 7/10, 1/2, 1/10)) %>% fractions()

## [1] 8/21  1/3 5/21 1/21
```

12

```
import re as re
re.findall(r'a+b*[cd]*', 'abcdabcddcbaabbccddccbbaa')
re.findall(r'b+a*[cd]*', 'abcdabcddcbaabbccddccbbaa')

## Error:  <text>:1:8:  unexpected symbol
## 1:  import re
##            ^
```

```
>>> import re as re
>>> re.findall(r'a+b*[cd]*', 'abcdabcddcbaabbccddccbbaa')
['abcd', 'abcddc', 'aabbccddcc', 'aa']
```

15 `(01 | 10*01*1)* 10*01*`